This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

Ultra-Early Prediction of Lithium-ion Battery Cycle Life Based on Visualized Single-Cycle Data

Wenjin Yang, *Graduate Student Member, IEEE,* Fanqi Min, Jingying Xie, and Hengzhao Yang, *Senior Member, IEEE*

***Abstract*—This paper proposes a battery cycle life prediction framework based on the visualized data of a single charging-discharging cycle during the ultra-early stage of the battery operation. To develop the framework, a sliding window-based image construction method is proposed that divides the raw sequential data extracted from a single cycle into multiple sub-sequences and uses the Euclidean distance between any two sub-sequences to construct the images. The framework employs three AlexNet blocks to build a sophisticated convolutional neural network model to capture the features from the images. Comprehensive evaluations of the framework are conducted using the Severson dataset [1] with 124 batteries. All the four models trained using the three measurements (i.e., voltage, current, and capacity) and the combination of them result in acceptably low cycle life prediction errors for the 29 batteries in the test set. Among the four models, the "Full" model based on the combination of the three measurements performs the best with an average root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) of 76.81 cycles, 7.05%, and 0.9178, respectively. As a feature-free method, the "Full" model outperforms three feature-based methods and another three feature-free methods, demonstrating its effectiveness in predicting the battery cycle life during the ultra-early stage with only the data of a single cycle.**

***Index Terms*—Lithium-ion battery, ultra-early cycle life prediction, single-cycle data, image construction, convolutional neural network.**

I. Introduction

Lithium-ion batteries have been widely adopted in electrified transportation systems and renewable energy systems due to their high energy density, low self-discharge rate, and long cycle life [1]–[4]. When the battery ages, its performance degrades and the safety risks increase. Therefore, various approaches have been proposed to estimate the battery state of health (SOH) [5]–[11]. Since the aging process is relatively slow, it is of great interest to predict the battery remaining useful life (RUL) during an early stage [1], [12]–[15]. Moreover, if the battery RUL can be accurately predicted using the data of a single charging-discharging cycle or a very limited number of cycles, the batteries can be sorted rapidly, which is particularly useful in accelerating the research and development of new batteries and recycling used batteries [16].

A variety of methods have been developed to predict the battery RUL, which can be roughly classified into two categories: model-based methods and data-driven methods [17]. Model-based methods usually rely on equivalent circuit models (ECMs) and electrochemical models (EMs) [17] of the battery. ECMs simulate the battery external characteristics by building circuits with electrical components (e.g., resistors and capacitors) [18]. The circuit components are usually not directly linked to the physical structures and processes of the battery [19]. EMs (e.g., pseudo-two-dimension (P2D) model) describe the fundamental mechanisms of the battery through a series of partial differential equations [19], [20]. For EMs, the number of parameters to be identified is usually large and the computational complexity of solving the partial differential equations is typically high.

On the other hand, data-driven methods are being extensively investigated to estimate the battery SOH and predict the battery RUL [21]–[27]. For applications in battery RUL prediction, data-driven methods can be roughly classified into three categories: time series forecasting (TSF) methods, feature-based methods, and feature-free methods. TSF methods directly utilize the historical capacity degradation trajectory to predict the future trajectory using popular tools such as recurrent neural network (RNN), long short-term memory (LSTM), gated recurrent unit (GRU), and Gaussian process regression (GPR) [28]–[33]. However, TSF methods generally need 40%-70% of the data to predict the future trajectory [34], which requires extensive experiments to generate sufficient data [35]. Correspondingly, the research and development cost of the RUL prediction method in this category is relatively high.

Feature-based methods utilize the features extracted from the raw data (i.e., voltage, current, and temperature) to train a machine learning (ML) or deep learning (DL) model to predict the battery RUL [36]–[39]. For example, features are extracted from the $Q(V)$ (i.e., relationship between capacity and voltage) curves of the first 100 cycles to train an ElasticNet model [1]. Features are also extracted from other data such as the discharging capacity, terminal voltage, discharging current, temperature, and electrochemical impedance spectroscopy (EIS) [40]–[43]. Feature-based methods usually

Corresponding author: Hengzhao Yang.

Wenjin Yang and Hengzhao Yang are with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China (email: yangwj2022@shanghaitech.edu.cn; hzyang@shanghaitech.edu.cn).

Fanqi Min is with the Shanghai Aerospace Power Technology Co., Ltd., Shanghai 201112, China and the State Key Laboratory of Space Power Source Technology, Shanghai Institute of Space Power-Sources, Shanghai 200245, China (email: mfq@sapt.cc).

Jingying Xie is with the State Key Laboratory of Space Power Source Technology, Shanghai Institute of Space Power-Sources, Shanghai 200245, China and the School of Chemistry and Chemical Engineering, Harbin Institute of Technology, Harbin 150001, China (email: jyxie@hit.edu.cn).

This work was supported in part by the National Natural Science Foundation of China, Beijing, China under Grant No. U22B2069 and in part by the Science and Technology Commission of Shanghai Municipality, Shanghai, China under Grant No. 22DZ1200600.

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

require expert knowledge to extract effective features related to the battery RUL. However, during the ultra-early stage of the battery operation, it is challenging to identify promising features because the degradation pattern is not obvious.

To reduce or even eliminate the dependence on the handcraft features, feature-free methods are being proposed. Convolutional neural networks (CNNs) are being introduced to predict the battery cycle life because of their capabilities of parameter sharing and automatic feature extraction [44]–[46]. For instance, the $Q(V)$ curves of the first 100 discharging cycles are constructed into images and a CNN model is trained to automatically extract features from the images [45]. In addition to the $Q(V)$ curves, feature curves including the incremental capacity curve and the capacity difference curve are utilized to construct the images [3]. A hybrid method that integrates CNN and LSTM (i.e., CNN-LSTM) is proposed to predict the battery RUL using the data of the first 30 cycles [47]. While these methods use the first 100 or fewer cycles to predict the battery cycle life, researchers have pushed the boundary to utilize the data of a single cycle. For instance, a one-dimensional CNN (1D-CNN) model based on the voltage, current, and temperature data of a single cycle is proposed to predict the battery degradation trajectory [44]. A sequence-to-sequence (Seq2Seq) model is utilized to predict the $Q(V)$ curves of the future cycles based on the $Q(V)$ curve of a single cycle [48]. However, these methods are all end-to-end methods and they usually require a large amount of data to train the models [3]. One challenge of developing such end-to-end methods is that battery testing is time consuming and it may take a long time to collect enough data [49]. For instance, the Severson dataset [1] only contains 124 batteries. For such a relatively small dataset, it may be challenging for the end-to-end methods to mine the complex and vital information of the data. To address this issue, one possibility is to develop more effective deep learning models for specific inputs [50]. For instance, the attention-assisted temporal convolutional memory-augmented network (ATCMN) is developed for battery RUL prediction with a limited amount of data [50]. A dilated CNN is employed to extract features from inter-cycle multi-dimensional data (i.e., terminal voltage, current, and temperature) [51].

Although certain models have been developed for ultra-early prediction of the battery cycle life, few studies have explored the capabilities of applying these models to single-cycle data of relatively small battery datasets. One attempt is to develop effective input structures (e.g., in the form of an image [3], [44], [45]) for deep learning models [50]. The images are usually constructed by stacking the one-dimensional sequential data in the height/depth direction. For single-cycle data, the images obtained in this fashion are still challenging for the CNNs to learn the effective patterns because of the relatively naive input structure. Therefore, this paper proposes a data preprocessing approach to transform the single-cycle sequential data into image-form data, which helps the CNN models understand the high-level patterns from the battery data despite the relatively small amount of the battery data. Additionally, this paper proposes a battery cycle life prediction framework based on the visualized data of a single cycle during the ultra-early stage of the battery operation. Specifically, this paper

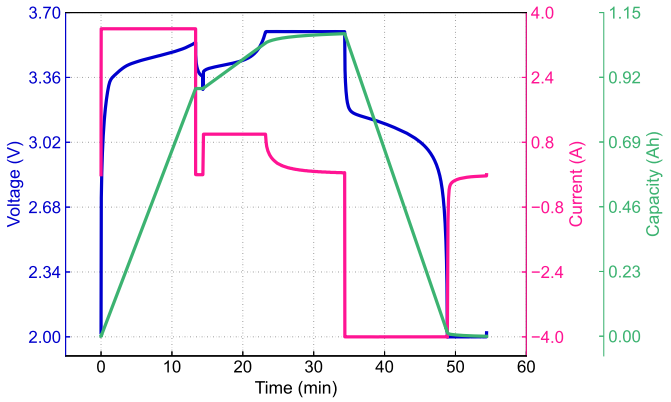


Fig. 1. Voltage, current, and capacity curves extracted from second cycle of "b1c1" cell.

makes three major contributions:

- First, an ultra-early battery cycle life prediction framework based on the visualized single-cycle data (i.e., voltage, current, and capacity) is proposed. Evaluation results using the Severson dataset [1] show that the average root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) of the "Full" model based on the combination of the voltage, current, and capacity data are 76.81 cycles, 7.05%, and 0.9178, respectively.
- Second, a sliding window-based image construction method is developed to transform the raw sequential data extracted from a single cycle into images. This method can not only avoid manual feature extraction but also capture the intrinsic relevance structure of the data. Therefore, the proposed image construction method can augment the data, enabling the deep learning models to better mine the patterns within the data.
- Third, a CNN model composed of three AlexNet [52] blocks is utilized to extract the features from the images.

The remainder of this paper is organized as follows. Section II describes the battery dataset used in this paper. Section III presents the ultra-early battery cycle life prediction framework. Section IV evaluates the proposed framework and demonstrates its effectiveness. Section V concludes this paper.

II. Battery Dataset

This paper uses the Severson dataset to develop and evaluate the proposed framework. The dataset contains 124 commercial LFP/graphite cells, which are divided into three batches: each with 41, 43, and 40 cells, respectively. These cells undergo charging-discharging cycling tests in a temperature chamber maintained at 30 °C. All the cells are rated with a nominal capacity of 1.1 Ah and a nominal voltage of 3.3 V.

For a complete charging-discharging cycle, each cell is first charged using a one-step or two-step fast-charging strategy until its state of charge (SOC) reaches 80%. The fast-charging strategy during this stage is denoted as "C1(Q1)-C2", where C1 and C2 are the currents during the two steps and Q1 is the SOC when the current changes from C1 to C2. If there is only one step, C1 and C2 are equal and Q1 is 80%. Next, the

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

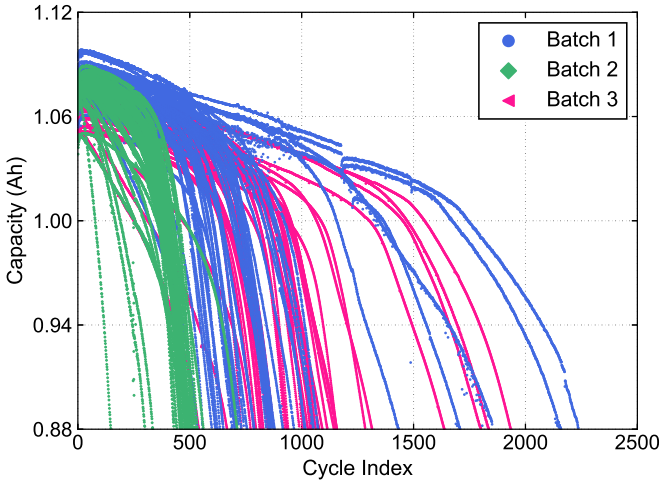


Fig. 2. Capacity degradation trajectories of all cells.

cell is charged at 1 C constant-current constant-voltage (CC-CV) until its SOC reaches 100%. The cut-off voltage of the charging process is set as 3.6 V. On the other hand, during the discharging process, the cell is discharged at a constant current of 4 C until its voltage reaches the discharging cut-off voltage of 2.0 V. Specifically, take the "b1c1" cell from batch 1 as an example. The voltage, current, and capacity curves extracted from its second complete charging-discharging cycle are shown in Fig. 1. The charging-discharging strategy is "3.6C(80%)-3.6C", which means that this cell is first charged in one-step at 3.6 C until its SOC reaches 80% SOC. The actual current is 3.6 A, which is approximately equal to the rate of 3.6 C (i.e., 3.6 C is 3.96 A when the nominal capacity is 1.1 Ah). After a short rest (i.e., 1.11 minutes), this cell is charged at 1 A CC-CV until its SOC reaches 100% and its voltage hits the upper bound of 3.6 V. Then, this cell is discharged at a constant current of -4 A until its voltage drops to the lower bound of 2.0 V. Note that for the cells in batch 1, the second cycle is the earliest cycle with usable test data because the sampling rate for the first cycle is too high and the test data of the first cycle is excluded from the dataset. For consistency, this paper extracts the data from the second cycle for all the cells in the three batches.

For the 124 cells in the Severson dataset, Fig. 2 shows their capacity degradation trajectories. The cycle life (i.e., the index of the cycle when the capacity decreases to 80% of the nominal capacity) of these cells varies between 148 and 2237 cycles, depending on the specific fast-charging policies employed to cycle them.

III. Battery Cycle Life Prediction Framework

A. Overview

The ultra-early battery cycle life prediction framework proposed in this paper is illustrated in Fig. 3. First, the voltage, current, and capacity curves are extracted from the second cycle of the battery test data. Next, the three curves are resized into the same length of L (e.g., $L = 1000$) and normalized into the range of [0,1]:

$$z_{\text{norm}} = \frac{z - z_{\min}}{z_{\max} - z_{\min}} \tag{1}$$

where z represents the raw data, z_{norm} represents the normalized data, $z_{\min}$ and $z_{\max}$ are the minimum and maximum values of the raw data. Third, the three normalized curves are transformed into the corresponding images. Fourth, the images with a size of $N \times N$ are resized into 224×224 to satisfy the size requirement of the CNN model utilized in this paper. Note that during the resizing process, if the width of the original image is less than 224, the resizing operation carries out bi-linear interpolation for the image. In contrast, if the width of the original image is greater than 224, the resizing operation carries out average pooling to down-sample the image. These three images constitute a three-channel image. Finally, the three-channel images and the cycle life labels are utilized to train a CNN model for ultra-early cycle life prediction.

B. Image Construction

CNNs are powerful in mining the local correlations in space. However, the global connections hidden in the sequential data are relatively difficult for CNNs to capture. To bridge this gap, this paper proposes a sliding window-based image construction method to capture the intrinsic relevance structure of the sequential data, as shown in Fig. 3.

The proposed image construction method uses a time window with a fixed window size (denoted as W) to move forward on the sequential data, dividing it into multiple sub-sequences. The number of the sub-sequences can be calculated as

$$N = \left\lfloor \frac{L - W}{S} \right\rfloor + 1 \tag{2}$$

where S is the stride. Then the set of the sub-sequences can be expressed as

$$\Phi = \{\mathbf{p}_i | i \in (1, N)\} \tag{3}$$

where $\mathbf{p}_i$ is the ith sub-sequence. To construct the image, the relationship matrix $\mathbf{F}$ between any two sub-sequences needs to be calculated, which can be defined as

$$\mathbf{F}_{i,j} = \|\mathbf{p}_i - \mathbf{p}_j\| \tag{4}$$

Therefore, $\mathbf{F}_{i,j}$ measures the Euclidean distance (i.e., degree of relevance) between $\mathbf{p}_i$ and $\mathbf{p}_j$. Then $\mathbf{F}$ is regarded as an $N \times N$-pixel image. As illustrated in Fig. 4, the corresponding images of the voltage, current, and capacity curves are displayed in Fig. 4(a), Fig. 4(b), and Fig. 4(c), respectively.

C. CNN Architecture

The architecture of the CNN model utilized in the proposed framework is shown in Fig. 3. AlexNet [52] is a deep CNN that can capture more complex features. Alexnet incorporates several key technologies, including the rectified linear unit (ReLU) activation function, the dropout regularization method, and local response normalization (LRN), which not only alleviate the gradient vanishing issue but also enhance the generalization performance and training speed of the model.

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

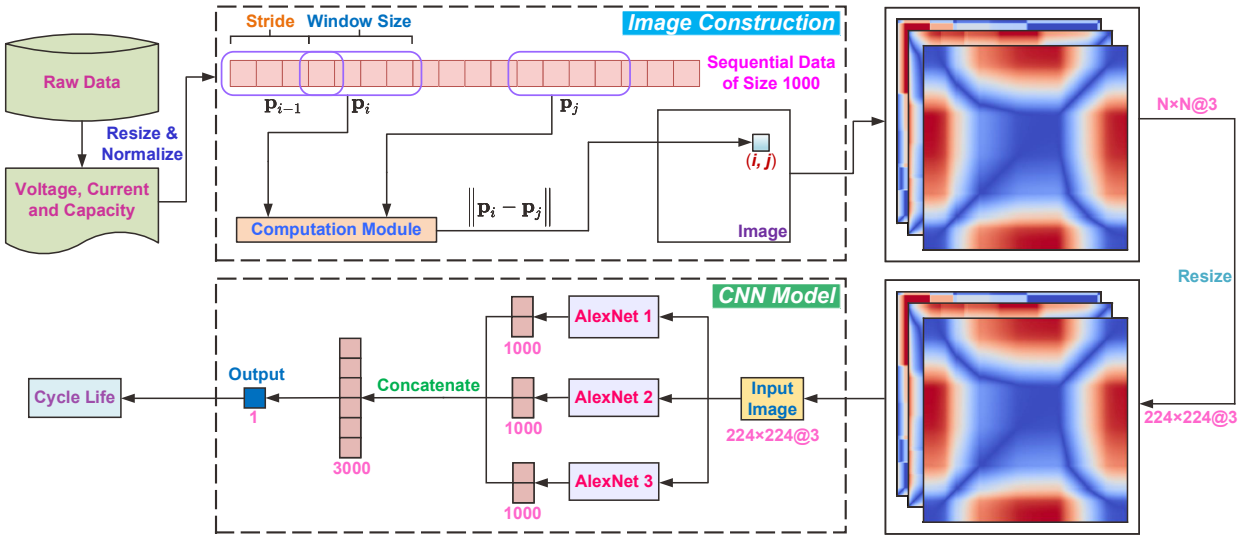


Fig. 3. Ultra-early battery cycle life prediction framework.

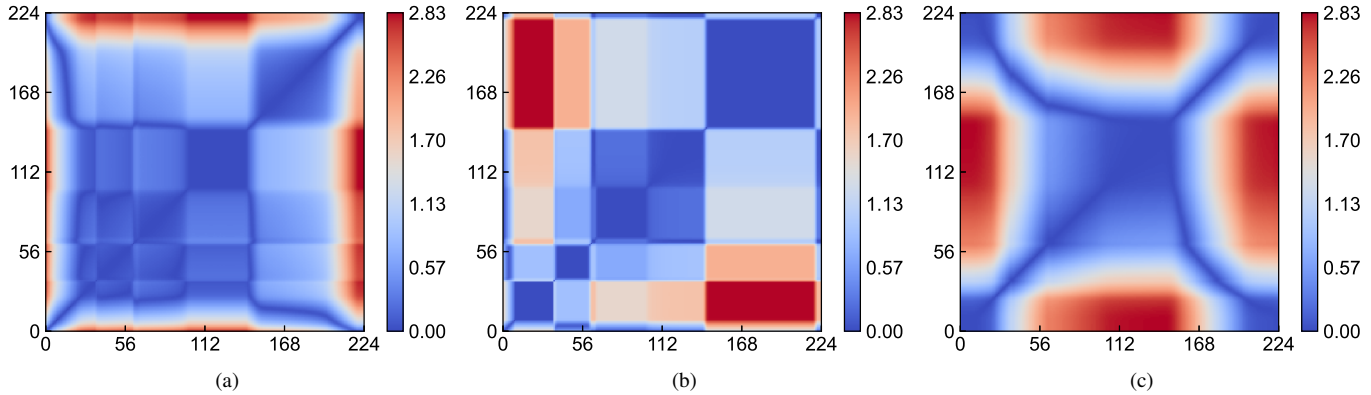


Fig. 4. Illustration of 224 × 224-pixel images obtained by applying proposed image construction method. (a) Voltage. (b) Current. (c) Capacity.

To effectively capture the features of multiple measurements (i.e., voltage, current, and capacity), the proposed CNN model consists of three AlexNet blocks. The output vector of each AlexNet includes 1000 neurons. The three output vectors are concatenated into a vector containing 3000 neurons, which is then fed into a fully connected layer.

IV. Evaluation of Proposed Framework

A. Evaluation Setup

The proposed framework is evaluated using the Severson dataset. Considering that the cycle life distribution of the batteries in the dataset is unbalanced, the dataset is partitioned as follows: (1) Out of the 11 batteries with a relatively long cycle life (i.e., no less than 1200 cycles), one battery is randomly selected and included in the test set. (2) All the three batteries with a relatively short cycle life (i.e., less than 400 cycles) are excluded from the test set. (3) 25% (27.5, rounded to 28) of the remaining 110 batteries are randomly selected and included in the test set. Therefore, there are 29 batteries in the test set. The remaining 95 batteries form the training and validation set.

The 19-fold cross-validation (CV) method is utilized for model selection (i.e., selection of the hyperparameters) in this paper. Specifically, the 95 batteries are partitioned into 19 mutually exclusive folds with each trial utilizing 90 batteries for training and 5 batteries for validation. This arrangement ensures that sufficient data can be retained for training. Moreover, it ensures a shorter training time compared to the leave-one-out CV (LOOCV) method because 19-fold CV conducts 19 trials while LOOCV conducts 95 trials. The model with the best performance across the 19 trials is selected and applied to the test set. To be clear, the entire process of dataset partitioning, model selection, and model testing is referred to as an experiment. For each model in this paper, a set of four experiments is performed to avoid contingencies. All the models in this paper are trained using the exactly same setup elaborated above. For different models, the experiments with the same index have the same dataset partition.

The mean square error (MSE) is selected as the loss function, which is determined as follows:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5}$$

where n is the number of batteries, y_i is the observed cycle life of battery i, and $\hat{y}_i$ is the predicted cycle life of battery i. The effectiveness of the proposed framework is evaluated using

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

three metrics: RMSE, MAPE, and R^2, which are determined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \tag{7}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{8}$$

where $\bar{y}$ is the average observed cycle life of the batteries.

B. Evaluation Results: Effects of Measurements

As illustrated in Fig. 1, the voltage, current, and capacity curves can be extracted from the battery test data. The three physical parameters (i.e., voltage, current, and capacity) are referred to as three "measurements" in the evaluation setup. Moreover, the combination of the three physical parameters is denoted as the fourth "measurement". This section first evaluates the effects of the measurements on the battery cycle life prediction results. Each of the four measurements is used to train a cycle life prediction model. For the three individual measurements (i.e., voltage, current, and capacity), the corresponding models are denoted by the respective measurements: "Voltage", "Current", and "Capacity". For the combined measurement, the model is denoted as the "Full" model. For each model, a set of four experiments is performed to avoid contingencies. The evaluation results for the batteries in the test set are listed in Table I. In addition to the evaluation results for each of the four experiments (i.e., "Experiment 1" through "Experiment 4"), the averages of the results are also calculated and listed in the "Average" row.

To train the models, the images are constructed in the following manner. For an individual measurement, the three-channel image is obtained by repeating the image corresponding to the individual measurement three times. The CNN model utilizes a single AlexNet. This is because an individual measurement contains less information and increasing the model complexity would worsen its generalization performance. For the combined measurement, the three-channel image is constructed by combining the images corresponding to the three individual measurements. The CNN model fuses three AlexNet blocks. In this case, the combined measurement contains more information, which requires a more complex model to capture additional features. The evaluation results in Table I are obtained by applying the models to the batteries in the test set. The models are the ones resulting in the lowest validation losses in Section IV-C. For a specific model, the validation loss varies when the stride and the window size change. Therefore, the strides and the window sizes of the models used in Table I are different. Specifically, for the "Voltage", "Current", "Capacity", and "Full" models, the strides and the window sizes are ($S = 9$, $W = 10$), ($S = 3$, $W = 10$), ($S = 3$, $W = 40$), and ($S = 1$, $W = 70$), respectively.

The evaluation results in Table I lead to two major observations. First, the proposed battery cycle life prediction framework is effective in that all the four models corresponding to the four measurements result in acceptably low prediction errors. In terms of the average MAPE, the "Voltage", "Current", "Capacity", and "Full" models result in an average MAPE of 7.45%, 12.55%, 9.73%, and 7.05%, respectively. Second, among the four models, the "Full" model performs the best, followed by the "Voltage" model with comparable performance, the "Capacity" model with poorer performance, and the "Current" model with the worst performance. This observation is valid in terms of all three metrics: RMSE, MAPE, and R^2. Physically, this observation seems reasonable as the "Full" model utilizes three measurements while each of the other three models only utilizes one individual measurement.

For all the four models, Fig. 5 shows the visualization results for Experiment 1. A general observation is that the "Full" model outperforms the other three models in predicting the cycle life of all the batteries in the test set. Graphically, the blue dots are closer to the diagonal in Fig. 5(d). In particular, for the battery with a relatively long cycle life (i.e., with an observed cycle life of 1638 cycles), Fig. 5(d) shows that the predicted cycle life of 1701.47 cycles is much closer to the observed cycle life compared to the predictions in Figs. 5(a)-5(c). The insets in Fig. 5 show the histograms of the prediction errors, which are calculated as the predicted cycle life minus the observed cycle life. As shown in Fig. 5(d), the "Full" model results in more batteries in the test set with a prediction error closer to zero. As for the maximum error (in terms of absolute value), the "Full" model results in 231.37 cycles, while the "Voltage", "Current", and "Capacity" models lead to 460.29, 460.16, and 443.97 cycles, respectively.

C. Evaluation Results: Effects of Strides and Window Sizes

To investigate the effects of the strides and the window sizes on the battery cycle life prediction performance, these two parameters are swept for the four models as follows: $S = 1, 3, 5, 7, 9$ and $W = 10, 20, 30, 40, 50, 60, 70, 80, 90, 100$. In each sweep, a set of four experiments is performed and the average error is calculated. Fig. 6 shows the average RMSE results for the four models in each sweep. Two observations can be made. First, a qualitative analysis of the RMSE patterns suggests that for the "Voltage" and "Capacity" models, the average RMSE increases in general when the window size increases for a specific stride. For the "Current" and "Full" models, however, no obvious patterns can be observed. Second, a quantitative analysis of the RMSE statistics shows that the "Full" model outperforms the other three models. Specifically, the bounds of the average RMSEs of the "Voltage", "Current", "Capacity", and "Full" models are [88.57, 127.71], [94.71, 130.08], [102.69, 159.83], and [67.85, 98.99] cycles, respectively. In terms of the range (i.e., the upper bound minus the lower bound) of the average RMSE, the "Full" model results in the smallest range of 31.14 cycles while the other three models result in larger ranges of 39.14, 35.37, and 57.14 cycles, respectively. In the meantime, the lower and upper bounds of the average RMSE of the "Full"

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

TABLE I
EFFECTS OF MEASUREMENTS ON EVALUATION RESULTS

	"Voltage" model			"Current" model			"Capacity" model			"Full" model		
	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2
Experiment 1	122.55	9.97%	0.8015	137.16	13.18%	0.7513	114.58	8.97%	0.8265	68.94	6.39%	0.9372
Experiment 2	62.20	7.01%	0.9511	107.79	11.76%	0.8533	164.28	12.58%	0.6593	67.94	6.51%	0.9417
Experiment 3	75.70	5.67%	0.9315	127.38	12.94%	0.8060	102.68	9.04%	0.8739	88.93	8.46%	0.9054
Experiment 4	93.81	7.14%	0.8497	147.97	12.32%	0.6261	95.30	8.32%	0.8449	81.44	6.83%	0.8867
Average	88.57	7.45%	0.8835	130.08	12.55%	0.7592	119.21	9.73%	0.8012	76.81	7.05%	0.9178

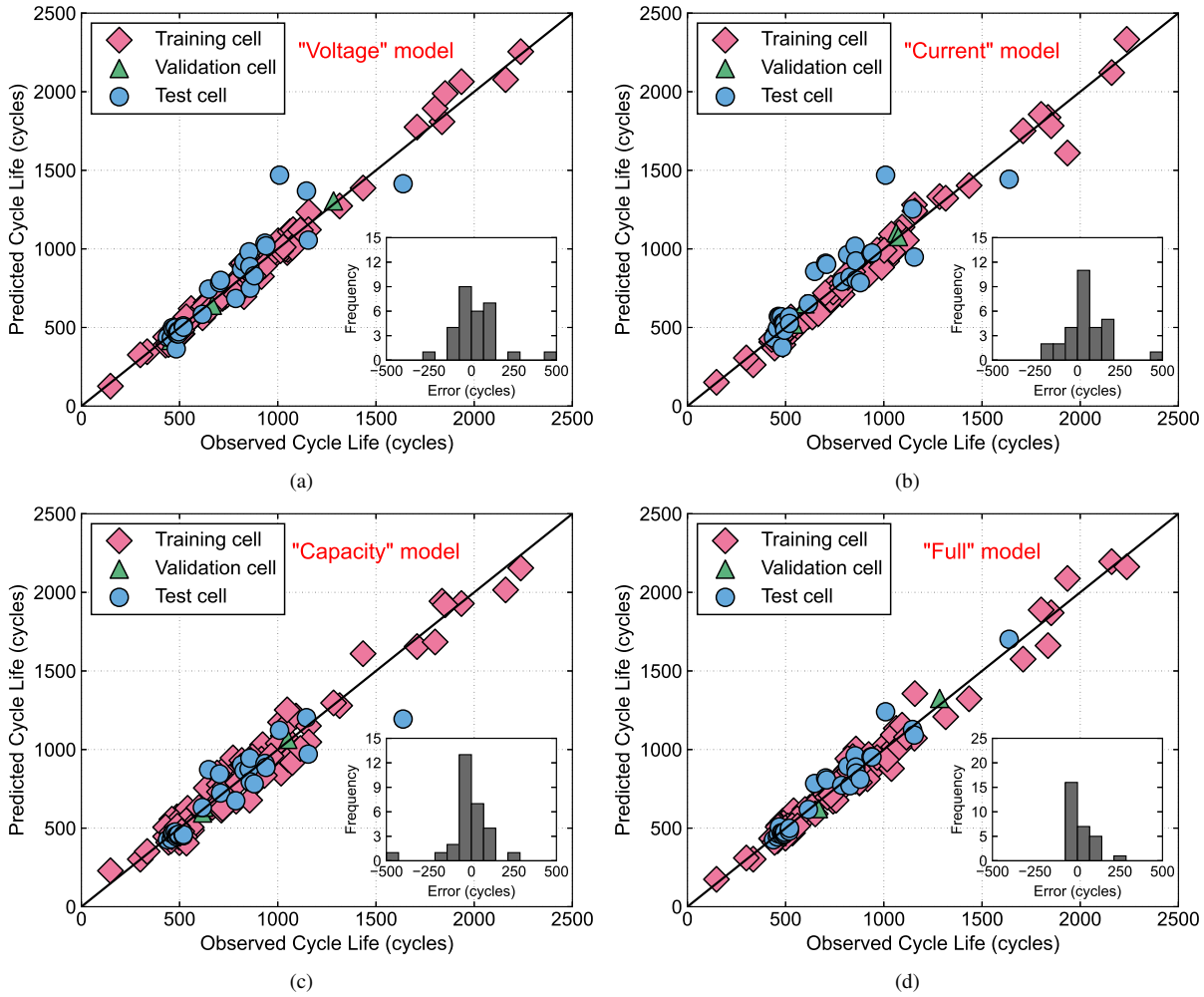


Fig. 5. Battery cycle life prediction results for Experiment 1. (a) "Voltage" model. (b) "Current" model. (c) "Capacity" model. (d) "Full" model.

model are also smaller than their counterparts of the other three models. In summary, Fig. 6 again shows that the "Full" model outperforms the other three models.

D. Evaluation Results: Comparison with Feature-based Methods

To demonstrate the superiority of the proposed framework over feature-based methods, three feature-based methods including ElasticNet, light gradient boosting machine (LightGBM), and extreme gradient boosting (XGBoost) are implemented using the exactly same setup elaborated in Section IV-B. The statistical features are extracted from the voltage, current, and capacity curves. To select the features that are closely related to the battery cycle life, Spearman correlation analysis [53] is conducted to evaluate the correlations between the battery cycle life and the extracted features.

To calculate the Spearman correlation coefficient, the following steps are executed. First, sort each set of data in ascending order and assign each data point x_i a ranking r_{x_i}. Second, calculate the Pearson correlation coefficient between the ranked features and the ranked battery cycle life label, which is the Spearman correlation coefficient:

$$\rho_x = \frac{\sum_{i=1}^{n}(r_{x_i} - \bar{r}_x)(r_{y_i} - \bar{r}_y)}{\sqrt{\sum_{i=1}^{n}(r_{x_i} - \bar{r}_x)^2 (r_{y_i} - \bar{r}_y)^2}} \tag{9}$$

where $\bar{r}_x$ and $\bar{r}_y$ are the means of the r_x and r_y sequences, r_{y_i} is the ith ranked battery cycle life label, and n is the

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

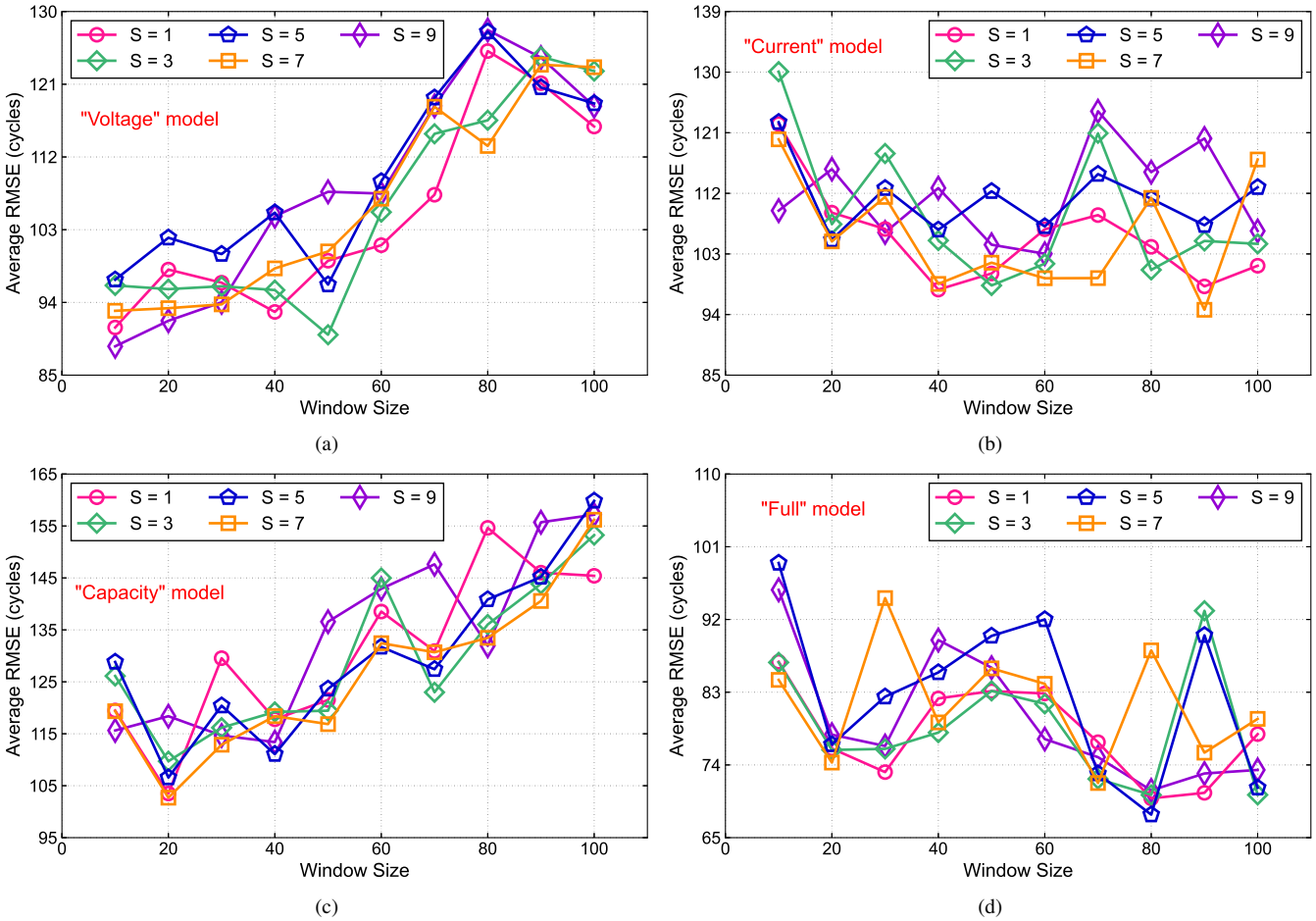


Fig. 6. Effects of strides and window sizes on average RMSE. (a) "Voltage" model. (b) "Current" model. (c) "Capacity" model. (d) "Full" model.

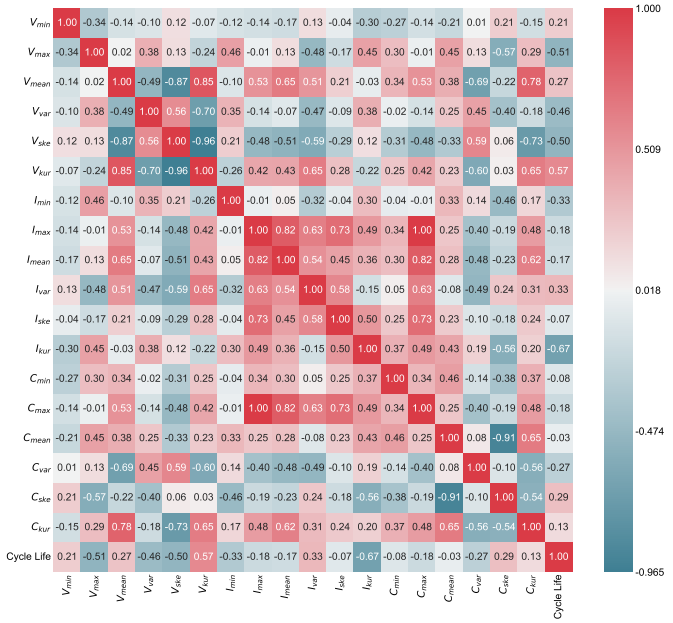


Fig. 7. Spearman correlation coefficients for 18 features.

number of the data samples. The extracted features include the maximum, minimum, mean, variance, skewness, and kurtosis of the voltage, current, and capacity curves (i.e., a total of 18 features). These statistical features reflect the data distribution characteristics of the three curves. For instance, skewness measures the asymmetry of the data distribution while kurtosis indicates the degree of the sharpness of the data distribution. The Spearman correlation coefficients between the 18 features and the battery cycle life are shown in Fig. 7. The features with an absolute correlation coefficient greater than 0.45 are selected, which include V_{max}, V_{var}, V_{ske}, V_{kur}, and I_{kur}. Their Spearman correlation coefficients are -0.51, -0.46, -0.50, 0.57, and -0.67, respectively. Note that although six features are extracted from the capacity curve, none of them shows an absolute correlation coefficient greater than 0.45. Therefore, none of them is selected.

The evaluation results of the three feature-based methods together with the "Full" model are shown in Fig. 8. Clearly, in terms of all the three metrics, the "Full" model performs better than all the three feature-based methods in all the four experiments. For instance, Fig. 8(b) shows that the average MAPEs of the "Full" model, ElasticNet, LightGBM, and XGBoost are 7.05%, 18.75%, 14.91%, and 13.00%, respectively. Among the three feature-based methods, XGBoost results in the minimum average MAPE of 13.00%, which is almost

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

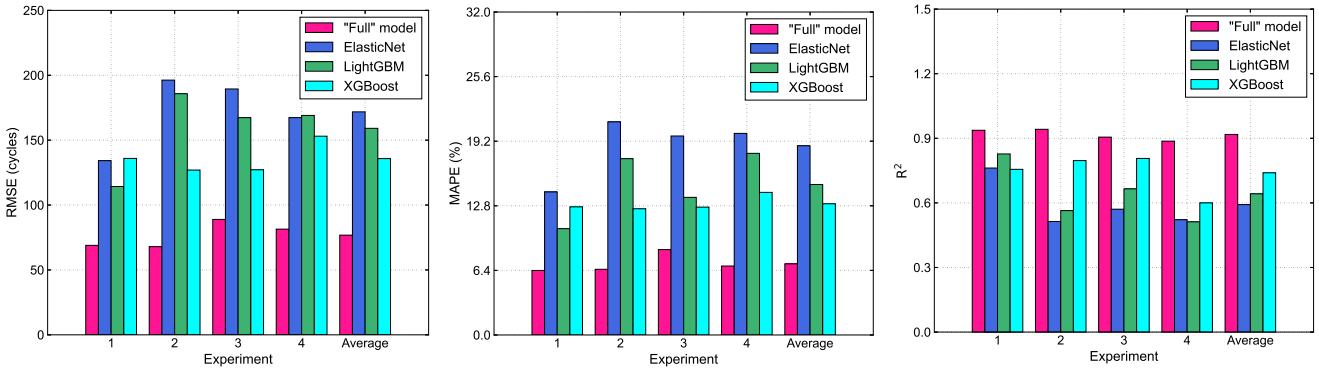


Fig. 8. Performance comparison between "Full" model and feature-based methods. (a) RMSE. (b) MAPE. (c) R^2.

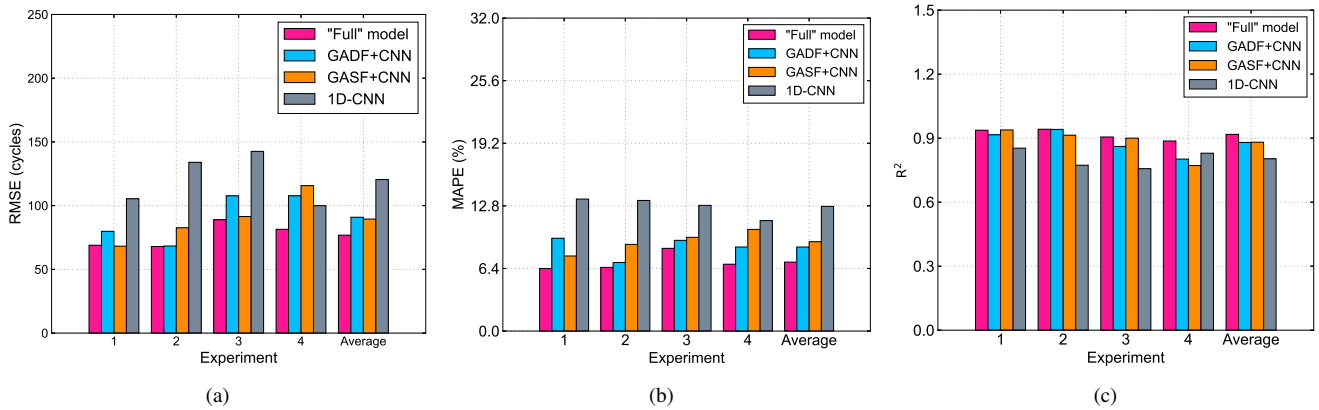


Fig. 9. Performance comparison between "Full" model and other feature-free methods. (a) RMSE. (b) MAPE. (c) R^2.

twice of the average MAPE of 7.05% given by the "Full" model. Note that the features utilized in the three feature-based methods are relatively naive. However, this also shows that extracting effective features related to the battery cycle life from the single-cycle data during the ultra-early stage is challenging.

E. Evaluation Results: Comparison with Other Feature-free Methods

To further demonstrate the effectiveness of the proposed framework, this section examines another three feature-free methods. Gramian angular difference/summation fields (GADF/GASF) can convert sequential data into images [54], which enables the use of CNNs to predict the battery cycle life. Therefore, another two image-based feature-free methods are proposed and investigated: "GADF+CNN" and "GASF+CNN". Note that the CNN architectures utilized in these two methods are the same as that used in the "Full" model. The third feature-free method is 1D-CNN, which is an end-to-end method with the raw sequential data (i.e., voltage, current, and capacity) as the model inputs. Again, the three feature-free methods are implemented using the exactly same setup (i.e., dataset partitioning and model selection) elaborated in Section IV-B. Note that the numbers of the needed parameters are 183.31 millions for the "Full" model , 183.31 millions for "GADF+CNN", 183.31 millions for "GASF+CNN", and 0.43 millions for 1D-CNN, respectively.

The evaluation results of the three feature-free methods together with the "Full" model are shown in Fig. 9. Two observations can be made. First, the "Full" model outperforms all the other three methods in almost all the four individual experiments. In terms of all the three metrics, the average results of the "Full" model are the best. For example, the average MAPEs of the "Full" model, "GADF+CNN", "GASF+CNN", and 1D-CNN are 7.05%, 8.59%, 9.13%, and 12.75%, respectively. For the individual experiments, the only scenario in which the performance of the "Full" model is marginally lower is Experiment 1. Specifically, the RMSE and R^2 of the "Full" model are very close to those of "GASF+CNN" (i.e., 68.94 versus 68.20 and 0.9372 versus 0.9385). Even in this scenario, the MAPE of the "Full" model is still lower than that of "GASF+CNN" (i.e., 6.39% versus 7.68%). For Experiment 1, the performance of the "Full" model is better than "GADF+CNN" and 1D-CNN. For Experiments 2-4, the "Full" model outperforms all the other three methods in terms of all the three metrics. Second, among the four feature-free methods including the proposed framework, 1D-CNN performs the worst, which suggests that converting the sequential data into images is more effective than end-to-end methods in battery cycle life prediction.

Finally, Fig. 10 compares the average RMSEs and MAPEs

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

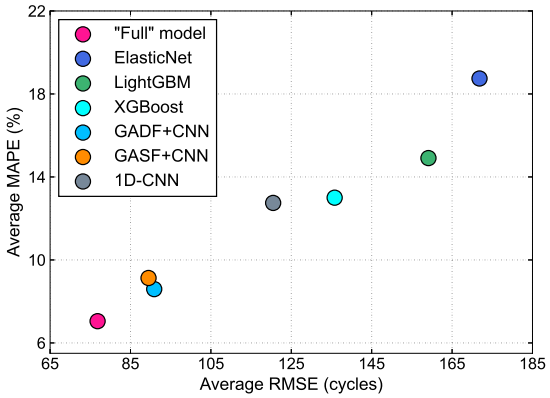


Fig. 10. Performance comparison of seven methods.

of all the seven methods. Clearly, the "Full" model trained using the proposed framework performs the best in terms of both RMSE and MAPE. Although not shown in the figure, the "Full" model also performs the best in terms of R^2. Between the two categories of methods, the four feature-free methods perform better than the three feature-based methods. For instance, the average RMSEs of ElasticNet, LightGBM, and XGBoost are 171.82, 159.12, and 135.79 cycles, respectively. For comparison, the average RMSEs of the "Full" model, "GADF+CNN", "GASF+CNN", and 1D-CNN are 76.81, 90.90, 89.49, and 120.49 cycles, respectively. Therefore, it may be more viable to develop feature-free methods for battery cycle life prediction during the ultra-early stage when only the data of a single cycle is utilized. Furthermore, among the four feature-free methods, the "Full" model still performs the best. Therefore, it demonstrates that the proposed image construction method is more effective than GADF and GASF in battery cycle life prediction during the ultra-early stage with the data of a single cycle.

F. Computation Burden Analysis

This section further analyzes the computation burdens of different image construction methods. Specifically, the computation process to be compared involves three steps. First, transform the sequential data (i.e., voltage, current, and capacity) of length $L = 1000$ into images. Second, resize these images into 224×224. Third, combine three images into a single three-channel image. The time consumed during this entire computation process is recorded as the computation time. The image construction methods are implemented using Python 3.10 on a computer with a 3.00 GHz Intel Core i5-12490F CPU and a 32 GB RAM.

For the proposed image construction method, to investigate the effects of the strides and the window sizes on the computation time, these two parameters are swept as follows: $S = 1, 3, 5, 7, 9$ and $W = 10, 20, 30, 40, 50, 60, 70, 80, 90, 100$. In each sweep, the average computation time for the 124 batteries in the dataset is recorded. Fig. 11(a) shows the average computation time in each sweep. Clearly, for a larger window size and a smaller stride, the average computation time is longer. For instance, the average computation time is 0.933

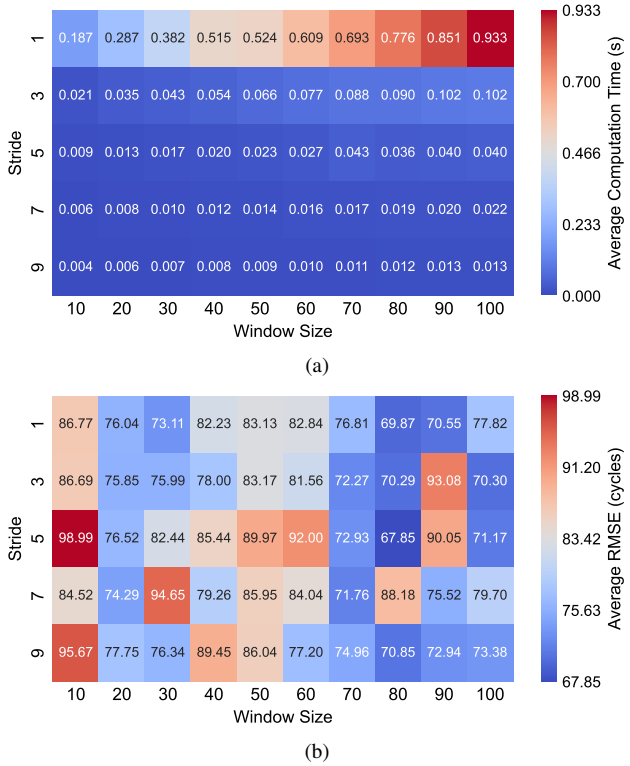


Fig. 11. Effects of strides and window sizes. (a) Average computation time of proposed image construction method. (b) Average RMSE of "Full" model.

s when $S = 1$ and $W = 100$. In contrast, the average computation time is only 0.004 s when $S = 9$ and $W = 10$.

For the other three feature-free methods examined in Section IV-E, the average computation time are also recorded. In particular, the average computation time for GADF and GASF are 0.098 and 0.096 s, respectively. Since the 1D-CNN method does not involve the data preprocessing stage, its average computation time is not recorded. For the "Full" model utilizing the proposed image construction method, Fig. 11(b) shows the average RMSEs when the stride and the window size vary. Referring to Section IV-E, the average RMSEs of "GADF+CNN" and "GASF+CNN" are 90.90 and 89.49 cycles, respectively. Therefore, between GADF and GASF, the latter performs better in terms of both the average computation time and the average RMSE, which is then used as the baseline in this section to evaluate the proposed image construction method. In particular, based on Fig. 11, the proposed method is evaluated against GASF in terms of the average computation time and the average RMSE of the corresponding battery cycle life prediction framework. Specifically, two comparisons can be performed to illustrate the benefits of the proposed method.

First, between GASF and the proposed method, when the average computation time are close, the proposed method results in a lower average RMSE. Given the limited resolutions (i.e., the sweep step sizes) of the stride and the window size, it is possible that none of the average computation time of the proposed method is exactly the same as that of GASF (i.e., 0.096 s). Therefore, the top three closest average computation time are extracted from Fig. 11(a). For $S = 3/W = 80$, $S = 3/W = 90$, and $S = 3/W = 100$, the

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

average computation time of the proposed method are 0.090, 0.102, and 0.102 s, respectively. Correspondingly, the average RMSEs are 70.29, 93.08, and 70.30 cycles, respectively, as shown in Fig. 11(b). While 93.08 is greater than 89.49, both 70.29 and 70.30 are lower than 89.49. Therefore, the proposed method results in a lower RMSE in general when its average computation time is close to that of GASF.

Second, between GASF and the proposed method, when the average RMSEs are close, the proposed method needs a shorter average computation time. Again, the top three average RMSEs closest to that of GASF (i.e., 89.49 cycles) are extracted. For $S = 9/W = 40$, $S = 5/W = 50$, and $S = 5/W = 90$, the average RMSEs are 89.45, 89.97, and 90.05 cycles, respectively. The corresponding average computation time are 0.008, 0.023, and 0.040 s, respectively, which are all significantly shorter than 0.096 s of GASF.

V. Conclusion

This paper proposes a sliding window-based image construction method in cooperation with AlexNet to predict the battery cycle life during the ultra-early stage. The proposed image construction method divides the raw sequential data into multiple sub-sequences and uses the Euclidean distance between two sub-sequences to measure the degree of relevance between them, which significantly enhances the capability of mining the information hidden in the raw sequential data. In addition, the AlexNet model effectively captures the features related to the battery cycle life from the images. Furthermore, the proposed method only utilizes the data of a single cycle to accurately predict the battery cycle life. The proposed method helps address the problem of accurately predicting the battery cycle life during the ultra-early stage with a very limited amount of data.

The proposed method is developed and evaluated based on the Severson dataset of 124 batteries. Main conclusions are as follows.

- First, all the four models trained using the three individual measurements (i.e., voltage, current, and capacity) extracted from the second cycle of the battery test data and the combination of them result in acceptably low cycle life prediction errors for the batteries in the test set.
- Second, compared to the "Voltage", "Current", and "Capacity" models utilizing only one individual measurement, the "Full" model utilizing the combination of the three individual measurements performs the best. For the evaluation setup with 95 batteries in the training and validation set and the remaining 29 batteries in the test set, the average RMSE, MAPE, and R^2 of the "Full" model are 76.81 cycles, 7.05%, and 0.9178, respectively.
- Third, the "Full" model trained using the proposed method outperforms the three feature-based methods (i.e., ElasticNet, LightGBM, and XGBoost) and the three feature-free methods (i.e., "GADF+CNN", "GASF+CNN", and 1D-CNN) examined in this paper in terms of the average RMSE, MAPE, and R^2. Furthermore, in terms of the computation time, the proposed image construction method is also advantageous when the window size and the stride are properly configured.
- Fourth, for ultra-early prediction of the battery cycle life using only the single-cycle data, the three feature-based methods perform worse than the four feature-free methods (i.e., the "Full" model, "GADF+CNN", "GASF+CNN", and 1D-CNN) because there is limited information within the single-cycle data and it is challenging to extract effective features related to the battery cycle life.
- Finally, among the four feature-free methods, the three methods based on raw sequential data augmentation (i.e., the "Full" model, "GADF+CNN", and "GASF+CNN") are more effective than the end-to-end method (i.e., 1D-CNN). The latter two conclusions provide insights for further development of the proposed framework.

References

[1] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H. Chen, M. Aykol, P. K. Herring, D. Fraggedakis, M. Z. Bazant, S. J. Harris, W. C. Chueh, and R. D. Braatz, "Data-driven prediction of battery cycle life before capacity degradation," *Nature Energy*, vol. 4, no. 5, pp. 383–391, 2019.

[2] F. Wang, Z. Zhai, B. Liu, S. Zheng, Z. Zhao, and X. Chen, "Open access dataset, code library and benchmarking deep learning approaches for state-of-health estimation of lithium-ion batteries," *Journal of Energy Storage*, vol. 77, p. 109884, 2024.

[3] N. He, Q. Wang, Z. Lu, Y. Chai, and F. Yang, "Early prediction of battery lifetime based on graphical features and convolutional neural networks," *Applied Energy*, vol. 353, p. 122048, 2024.

[4] A. Tang, Y. Xu, Y. Hu, J. Tian, Y. Nie, F. Yan, Y. Tan, and Q. Yu, "Battery state of health estimation under dynamic operations with physics-driven deep learning," *Applied Energy*, vol. 370, p. 123632, 2024.

[5] X. Hu, L. Xu, X. Lin, and M. Pecht, "Battery lifetime prognostics," *Joule*, vol. 4, no. 2, pp. 310–346, 2020.

[6] X. Bian, Z. Wei, W. Li, J. Pou, D. U. Sauer, and L. Liu, "State-of-health estimation of lithium-ion batteries by fusing an open circuit voltage model and incremental capacity analysis," *IEEE Transactions on Power Electronics*, vol. 37, no. 2, pp. 2226–2236, 2022.

[7] J. Lu, R. Xiong, J. Tian, C. Wang, and F. Sun, "Deep learning to estimate lithium-ion battery state of health without additional degradation experiments," *Nature Communications*, vol. 14, no. 1, p. 2760, 2023.

[8] D. Zhang, Z. Wang, P. Liu, C. She, Q. Wang, L. Zhou, and Z. Qin, "A multi-step fast charging-based battery capacity estimation framework of real-world electric vehicles," *Energy*, vol. 294, p. 130773, 2024.

[9] X. Li, D. Yu, S. B. Vilsen, and D. I. Stroe, "Accuracy comparison and improvement for state of health estimation of lithium-ion battery based on random partial recharges and feature engineering," *Journal of Energy Chemistry*, vol. 92, pp. 591–604, 2024.

[10] S. Peng, Y. Sun, D. Liu, Q. Yu, J. Kan, and M. Pecht, "State of health estimation of lithium-ion batteries based on multi-health features extraction and improved long short-term memory neural network," *Energy*, vol. 282, p. 128956, 2023.

[11] X. Sui, S. He, and R. Teodorescu, "Small-sample-learning-based lithium-ion batteries health assessment: An optimized ensemble framework," *IEEE Transactions on Industry Applications*, vol. 60, no. 3, pp. 4366–4380, 2024.

[12] B. Cui, H. Wang, R. Li, L. Xiang, H. Zhao, R. Xiao, S. Li, Z. Liu, G. Yin, X. Cheng, Y. Ma, H. Huo, P. Zuo, T. Lu, J. Xie, and C. Du, "Ultra-early prediction of lithium-ion battery performance using mechanism and data-driven fusion model," *Applied Energy*, vol. 353, p. 122080, 2024.

[13] X. Tang, X. Lai, C. Zou, Y. Zhou, J. Zhu, Y. Zheng, and F. Gao, "Detecting abnormality of battery lifetime from first-cycle data using few-shot learning," *Advanced Science*, vol. 11, no. 6, p. 2305315, 2024.

[14] W. Li, N. Sengupta, P. Dechent, D. Howey, A. Annaswamy, and D. U. Sauer, "One-shot battery degradation trajectory prediction with deep learning," *Journal of Power Sources*, vol. 506, p. 230024, 2021.

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791

[15] J. Meng, L. Cai, S. Yang, J. Li, F. Zhou, J. Peng, and Z. Song, "An empirical-informed model for the early degradation trajectory prediction of lithium-ion battery," *IEEE Transactions on Energy Conversion*, vol. 39, no. 4, pp. 2299–2311, 2024.

[16] M. Zhao, Y. Zhang, and H. Wang, "Battery degradation stage detection and life prediction without accessing historical operating data," *Energy Storage Materials*, vol. 69, p. 103441, 2024.

[17] K. Liu, Y. Shang, Q. Ouyang, and W. D. Widanage, "A data-driven approach with uncertainty quantification for predicting future capacities and remaining useful life of lithium-ion battery," *IEEE Transactions on Industrial Electronics*, vol. 68, no. 4, pp. 3170–3180, 2021.

[18] D. Bodnár, G. R. C. Mouli, F. Ďurovský, P. Bauer, and Z. Qin, "Semi-empirical model of nickel manganese cobalt (NMC) lithium-ion batteries including capacity regeneration phenomenon," *IEEE Transactions on Transportation Electrification*, in press.

[19] W. Li, L. Liang, W. Liu, and X. Wu, "State of charge estimation of lithium-ion batteries using a discrete-time nonlinear observer," *IEEE Transactions on Industrial Electronics*, vol. 64, no. 11, pp. 8557–8565, 2017.

[20] J. Sturm, A. Rheinfeld, I. Zilberman, F. Spingler, S. Kosch, F. Frie, and A. Jossen, "Modeling and simulation of inhomogeneities in a 18650 nickel-rich, silicon-graphite lithium-ion cell during fast charging," *Journal of Power Sources*, vol. 412, pp. 204–223, 2019.

[21] Z. Wei, H. Ruan, Y. Li, J. Li, C. Zhang, and H. He, "Multistage state of health estimation of lithium-ion battery with high tolerance to heavily partial charging," *IEEE Transactions on Power Electronics*, vol. 37, no. 6, pp. 7432–7442, 2022.

[22] Z. Deng, X. Hu, P. Li, X. Lin, and X. Bian, "Data-driven battery state of health estimation based on random partial charging data," *IEEE Transactions on Power Electronics*, vol. 37, no. 5, pp. 5021–5031, 2022.

[23] V. Steininger, K. Rumpf, P. Hüsson, W. Li, and D. U. Sauer, "Automated feature extraction to integrate field and laboratory data for aging diagnosis of automotive lithium-ion batteries," *Cell Reports Physical Science*, vol. 4, no. 10, p. 101596, 2023.

[24] J. Zhu, Y. Wang, Y. Huang, R. B. Gopaluni, Y. Cao, M. Heere, M. J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, A. Senyshyn, X. Wei, M. Knapp, and H. Ehrenberg, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation," *Nature Communications*, vol. 13, no. 1, p. 2261, 2022.

[25] Q. Li, J. Zhong, J. Du, Y. Yi, J. Tian, Y. Li, C. Lai, T. Lu, and J. Xie, "Probabilistic neural network-based flexible estimation of lithium-ion battery capacity considering multidimensional charging habits," *Energy*, vol. 294, p. 130881, 2024.

[26] Y. Xu and H. Yang, "A battery capacity trajectory prediction framework with mileage correction for electric buses," *Journal of Energy Storage*, vol. 110, p. 115301, 2025.

[27] Y. Xu, Y. Zhang, W. Yang, C. Xu, F. Wang, and H. Yang, "A battery state of health prediction framework considering user behavior for on-road electric vehicles," in *Proceedings of the 2024 IEEE Transportation Electrification Conference and Exposition (ITEC 2024)*, 2024, pp. 1–6.

[28] Z. Wang, N. Liu, C. Chen, and Y. Guo, "Adaptive self-attention LSTM for RUL prediction of lithium-ion batteries," *Information Sciences*, vol. 635, pp. 398–413, 2023.

[29] Y. Zhang, R. Xiong, H. He, and M. G. Pecht, "Long short-term memory recurrent neural network for remaining useful life prediction of lithium-ion batteries," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 7, pp. 5695–5705, 2018.

[30] F. Guo, X. Wu, L. Liu, J. Ye, T. Wang, L. Fu, and Y. Wu, "Prediction of remaining useful life and state of health of lithium batteries based on time series feature and Savitzky-Golay filter combined with gated recurrent unit neural network," *Energy*, vol. 270, p. 126880, 2023.

[31] L. Li, Y. Li, R. Mao, L. Li, W. Hua, and J. Zhang, "Remaining useful life prediction for lithium-ion batteries with a hybrid model based on TCN-GRU-DNN and dual attention mechanism," *IEEE Transactions on Transportation Electrification*, vol. 9, no. 3, pp. 4726–4740, 2023.

[32] Y. Wang, C. Hei, H. Liu, S. Zhang, and J. Wang, "Prognostics of remaining useful life for lithium-ion batteries based on hybrid approach of linear pattern extraction and nonlinear relationship mining," *IEEE Transactions on Power Electronics*, vol. 38, no. 1, pp. 1054–1063, 2023.

[33] W. Guo, Z. Sun, Y. Li, S. Jin, S. B. Vilsen, and D. I. Stroe, "Health status estimation for lithium-ion batteries with partial charging information using mixed inputs LSTM," in *Proceedings of the 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)*, 2024, pp. 1673–1679.

[34] J. Hu and L. Wu, "Early uncertainty quantification prediction of lithium-ion battery remaining useful life with transformer ensemble model," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 4, pp. 8618–8629, 2024.

[35] X. Jia, C. Zhang, Y. Li, C. Zou, L. Y. Wang, and X. Cai, "Knee-point-conscious battery aging trajectory prediction based on physics-guided machine learning," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, pp. 1056–1069, 2024.

[36] Z. Fei, F. Yang, K.-L. Tsui, L. Li, and Z. Zhang, "Early prediction of battery lifetime via a machine learning based framework," *Energy*, vol. 225, p. 120205, 2021.

[37] Z. Zhou, Y. Liu, M. You, R. Xiong, and X. Zhou, "Two-stage aging trajectory prediction of LFP lithium-ion battery based on transfer learning with the cycle life prediction," *Green Energy and Intelligent Transportation*, vol. 1, no. 1, p. 100008, 2022.

[38] H. Zhang, Y. Su, F. Altaf, T. Wik, and S. Gros, "Interpretable battery cycle life range prediction using early cell degradation data," *IEEE Transactions on Transportation Electrification*, vol. 9, no. 2, pp. 2669–2682, 2023.

[39] W. Han and C. Geng, "An accurate battery state of health estimation method easy to imlement after charging," *iEnergy*, vol. 2, no. 4, pp. 257–257, 2023.

[40] Y. Zhang, Z. Peng, Y. Guan, and L. Wu, "Prognostics of battery cycle life in the early-cycle stage based on hybrid model," *Energy*, vol. 221, p. 119901, 2021.

[41] F. Yang, D. Wang, F. Xu, Z. Huang, and K.-L. Tsui, "Lifespan prediction of lithium-ion batteries based on various extracted features and gradient boosting regression tree model," *Journal of Power Sources*, vol. 476, p. 228654, 2020.

[42] T. Li, Z. Zhou, A. Thelen, D. A. Howey, and C. Hu, "Predicting battery lifetime under varying usage conditions from early aging data," *Cell Reports Physical Science*, vol. 5, no. 4, p. 101891, 2024.

[43] R. Liu, D. Zhang, Z. Li, P. Bauer, and Z. Qin, "Feature extraction from electrochemical impedance spectroscopy for state of health estimation of lithium-ion batteries under different temperatures," in *Proceedings of the 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)*, 2024, pp. 3374–3378.

[44] S. Kim, H. Jung, M. Lee, Y. Y. Choi, and J.-I. Choi, "Model-free reconstruction of capacity degradation trajectory of lithium-ion batteries using early cycle data," *eTransportation*, vol. 17, p. 100243, 2023.

[45] S. Saxena, L. Ward, J. Kubal, W. Lu, S. Babinec, and N. Paulson, "A convolutional neural network model for battery capacity fade curve prediction using early life data," *Journal of Power Sources*, vol. 542, p. 231736, 2022.

[46] C.-W. Hsu, R. Xiong, N.-Y. Chen, J. Li, and N.-T. Tsou, "Deep neural network battery life and voltage prediction by using data of one cycle only," *Applied Energy*, vol. 306, p. 118134, 2022.

[47] S. Xu, X. Yang, G. Ma, and Y. Yuan, "RMDA: A regressive multiple-source domain adaption approach for early prediction of lithium-ion battery lifetime," in *Proceedings of the 2023 42nd Chinese Control Conference (CCC 2023)*, 2023, pp. 6981–6986.

[48] J. Tian, R. Xiong, W. Shen, and J. Lu, "Data-driven battery degradation prediction: Forecasting voltage-capacity curves using one-cycle data," *EcoMat*, vol. 4, no. 5, p. e12213, 2022.

[49] T. Han, Z. Wang, and H. Meng, "End-to-end capacity estimation of lithium-ion batteries with an enhanced long short-term memory network considering domain adaptation," *Journal of Power Sources*, vol. 520, p. 230823, 2022.

[50] Z. Fei, Z. Zhang, F. Yang, and K.-L. Tsui, "A deep attention-assisted and memory-augmented temporal convolutional network based model for rapid lithium-ion battery remaining useful life predictions with limited data," *Journal of Energy Storage*, vol. 62, p. 106903, 2023.

[51] J. Hong, D. Lee, E.-R. Jeong, and Y. Yi, "Towards the swift prediction of the remaining useful life of lithium-ion batteries with end-to-end deep learning," *Applied Energy*, vol. 278, p. 115646, 2020.

[52] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017.

[53] Y. Zhou, M. Huang, Y. Chen, and Y. Tao, "A novel health indicator for on-line lithium-ion batteries remaining useful life prediction," *Journal of Power Sources*, vol. 321, pp. 1–10, 2016.

[54] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," in *Proceedings of the 24th International Conference on Artificial Intelligence (IJCAI 2015)*, 2015, pp. 3939–3945.

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPEL.2025.3531791



Wenjin Yang (S'23) received the B.S. degree from ShanghaiTech University, Shanghai, China, in 2022. He is working toward the M.S. degree at ShanghaiTech University. His research interests include the state of health (SOH) estimation and remaining useful life (RUL) prediction for batteries.



Fanqi Min received the M.S. degree in chemical and environmental engineering from the Shanghai Institute of Technology, Shanghai, China, in 2015. He is currently a Ph.D. student with the Harbin Institute of Technology, Harbin, China, and a Research Engineer with the Shanghai Aerospace Power Technology Co., Ltd., Shanghai, China. His research interests include the state of health (SOH) estimation, remaining useful life (RUL) prediction, and failure analysis for batteries.



Jingying Xie received the Ph.D. degree in chemical engineering from the Harbin Institute of Technology, Harbin, China, in 1997. Since 2010, she has been a Chief Scientist with the Shanghai Institute of Space Power-Sources, Shanghai, China and an Adjunct Professor with the School of Chemistry and Chemical Engineering, Harbin Institute of Technology. Her research interests include the design, development, and application of lithium-based batteries and energy storage systems.



Hengzhao Yang (M'13-SM'20) received the B.S. degree in optoelectronics from Chongqing University, Chongqing, China, the M.S. degree in microelectronics and solid-state electronics from the Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai, China, and the Ph.D. degree in electrical and computer engineering from the Georgia Institute of Technology, Atlanta, GA, USA, respectively. He is an Assistant Professor with ShanghaiTech University, Shanghai, China. His research interests include energy storage technologies (e.g., supercapacitors, batteries, and hydrogen) for applications in electrified transportation systems and renewable energy systems. He is an Associate Editor for the IEEE Transactions on Industry Applications and a Track Chair for several IEEE conferences including APEC, ECCE, IECON, ITEC, and PEDG.

Authorized licensed use limited to: ShanghaiTech University. Downloaded on January 21,2025 at 12:16:02 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.